

Response to Reviews

Manuscript ID: TCSVT-26162-2025.R1

Title: Pose-Guided Multi-Cue Explicit Query Construction for Disambiguating Human-Object Interactions

Dear Editor and Reviewers,

Thank you very much for your valuable feedback and constructive suggestions. We sincerely appreciate the time and effort of the Editor-in-Chief, Associate Editor, Senior Area Editor, and all reviewers. Following your comments, we have thoroughly revised the manuscript and provided detailed responses below.

For clarity, the comments are in black, our responses are highlighted in blue, and the corresponding page numbers and paper screenshots are marked in orange.

Response to Editor-in-Chief

Based on the enclosed set of reviews, I am recommending that the manuscript undergo a **MINOR REVISION** and be resubmitted for consideration by the IEEE Transactions on Circuits and Systems for Video Technology.

REPLY:

We thank the Editor-in-Chief for this recommendation. We have revised the manuscript and responded to the Associate Editor, Senior Area Editor, and Reviewers' comments to improve the quality of the manuscript.

We understand that the reason why you select the IEEE TCSVT for your manuscript is that your manuscript has a good match with this journal---Many related papers should have already been published in this journal. Therefore, before your new submission, you have to answer two questions clearly in your revised manuscript and responses: a) what are the 3-5 papers published in the IEEE Transactions on Circuits and Systems for Video Technology, which are most closely related to your manuscript; b) what is distinctive / new about your current manuscript related to these previously published papers.

REPLY:

We sincerely thank the Editor-in-Chief for the valuable guidance.

As requested, we carefully examined the recent publications in IEEE TCSVT and identified several works that are most closely related to our manuscript. Below, we list the representative papers and explain how our work differs from them.

(a) Related papers:

[1] RR-Net: Relation Reasoning for End-to-End Human-Object Interaction Detection[J]. IEEE

Transactions on Circuits and Systems for Video Technology, 2021, 32(6): 3853-3865. [\[cited in the first paragraph of Section II.A on page 2\]](#)

[2] Multi-Scale Human-Object Interaction Detector[J]. IEEE Transactions on Circuits and Systems for Video Technology, 2022, 33(4): 1827-1838. [\[cited in the first paragraph of Section II.A on page 2\]](#)

[3] Exploring Spatio-Temporal Graph Convolution for Video-Based Human-Object Interaction Recognition[J]. IEEE Transactions on Circuits and Systems for Video Technology, 2023, 33(10): 5814-5827. [\[cited in the second paragraph of Section III.B on page 4\]](#)

[4] TED-Net: Dispersal Attention for Perceiving Interaction Region in Indirectly-Contact HOI Detection[J]. IEEE Transactions on Circuits and Systems for Video Technology, 2024, 34(7): 5603-5615. [\[cited in the third paragraph of Section I on page 1\]](#)

[5] ASCFormer: An Adaptive Structure-aware Cascaded Transformer for 3D Object Detection. IEEE Transactions on Circuits and Systems for Video Technology, 2025, *Early Access*. [\[cited in the second paragraph of Section II.B on page 3\]](#)

(b): Our innovations:

Compared with these related papers, our work is distinctive in explicitly constructing pose-guided multi-cue queries that jointly encode appearance, spatial, and fine-grained pose information through a hierarchical pose contextualization mechanism. Furthermore, a two-stage context-aware reasoning scheme enables more discriminative and robust interaction understanding without relying on textual priors. These design choices make our framework both conceptually and practically different from previous methods published in IEEE TCSVT.

Response to Associate Editor

All reviewers show positive comments since most issues have been well addressed. But they also consider that there are some minor problems should be further refined, e.g, missing references and typos.

REPLY:

Thank you for the Associate Editor's positive comments. We have carefully revised the manuscript to address the remaining minor issues, including correcting typos and completing the reference list. These refinements further improve the clarity and presentation of the paper.

Response to Reviewer 1

Comment 1:

The main motivation of the paper is to improve the performance of HOI by utilizing fine-grained information of human body posture. However, the details of modeling human body posture information in the paper are not detailed enough, and the ablation experiments related to human body posture are not sufficient. It is suggested to further improve it.

REPLY:

Thank you for this insightful comment. We agree that the posture modeling process required

clearer and more explicit descriptions.

1. Clarification of Fine-Grained Human Posture Modeling.

In the revised manuscript, we have substantially reorganized and expanded the P-CR branch to provide a detailed and transparent explanation of how human body posture is modeled and utilized for HOI recognition.

Specifically, we made the following improvements:

- (1) Structural clarification: We reformulated the P-CR branch as a clearly defined three-stage pipeline, including (i) coarse-grained global posture encoding, (ii) fine-grained local joint geometry modeling, and (iii) feature fusion, which explicitly reveals the information flow from pose estimation to interaction reasoning. The revised description unifies motivation, illustrative examples, and algorithmic details, thereby providing a clearer and more comprehensive explanation of the posture modeling process.
- (2) Explicit fine-grained posture modeling: We detailed the construction of joint-level geometric descriptors by computing relative distances and orientation angles between adjacent joints along the predefined kinematic structure (Fig.~5(a)), thereby clarifying what constitutes fine-grained posture information.
- (3) Context-aware joint selection: We further clarified the role of the Global Pose-Modulated Local Attention (GPLA) mechanism, explaining how global posture embeddings guide the weighting of interaction-relevant joints to suppress noise and enhance discriminative motion patterns.

Below is our revised content [P-CR Branch in Section III.B on pages 5-6]:

P-CR Branch. This branch is designed to explicitly model human body posture at two complementary granularities, enabling robust disambiguation of semantically similar HOIs [3], [6]. Rather than directly relying on raw pose estimator outputs, the proposed P-CR branch decomposes posture representation into three sequential components: (1) coarse-grained global posture encoding to capture holistic body configurations, (2) fine-grained local joint geometry modeling to characterize subtle joint-level variations, and (3) a feature fusion stage that integrates global and local cues into a unified posture embedding. This hierarchical design reflects the observation that different HOIs may depend on posture cues at distinct levels of granularity.

(1) Global posture encoding

Global posture encodes the overall configuration of the human body, including balance, stability, and coordinated movement across major body parts [34]. Such holistic information captures how joints are jointly organized during an interaction and provides a high-level description of action morphology. For instance, as illustrated in Fig. 5(b), the “eyepiece imaging” action requires coordinated full-body engagement: the head moves toward the tube, the hands stabilize the instrument, and the torso leans forward while maintaining a seated posture. These characteristics are difficult to infer from isolated joint patterns alone, yet are well reflected by global posture cues.

Specifically, we first concatenate the 18 keypoints of each human instance according to the fixed ordering shown in Fig. 5(a). The concatenated pose representation is then flattened and mapped into a compact embedding space, yielding the global posture features $\mathbf{V}_{pg} \in \mathbb{R}^{N_p \times D_{pg}}$:

$$\mathbf{V}_{pg} = f_{pg}(\text{reshape}(\mathbf{V}_p, (N_p, -1))). \quad (6)$$

where f_{pg} is a Feed-Forward Network (FFN).

(2) Local joint geometry modeling

While global posture captures the overall morphology of an interaction, many HOI categories differ primarily in fine-grained joint configurations [35]. As shown in Fig. 5(c), the “adjusting the limit screw” action is largely determined by precise arm and hand configurations, whereas other body parts may exhibit substantial inter-subject variability.

To explicitly model such fine-grained posture information, we derive structured geometric descriptors from raw pose features. Specifically, we compute Euclidean distances $\mathbf{V}_{pl,d} \in \mathbb{R}^{N_p \times 17}$ and orientation angles $\mathbf{V}_{pl,a} \in \mathbb{R}^{N_p \times 17}$ between local joints:

$$\mathbf{V}_{pl,d}, \mathbf{V}_{pl,a} = \mathcal{G}(\mathbf{V}_p), \quad (7)$$

where $\mathcal{G}(\cdot)$ denotes a geometric mapping that converts raw pose features $\mathbf{V}_p$ into relative joint-level representations, following the fixed node ordering and kinematic structure shown in Fig. 5(a).

Direct aggregation of local joint features may introduce noise, as not all joints contribute equally to a given interaction. To address this issue, we introduce a Global Pose-Modulated Local Attention (GPLA) mechanism, in which coarse-grained global posture guides the selection of interaction-relevant joints. The key intuition is that global posture embeddings capture high-level action morphology and execution patterns, which provide informative priors for weighting local joints. By conditioning joint-level aggregation on these global cues, GPLA emphasizes interaction-relevant joints while suppressing interaction-irrelevant ones, enabling more accurate modeling of subtle motion patterns for HOI disambiguation [36].

Formally, joint-wise attention weights $\mathbf{A}_p \in \mathbb{R}^{N_p \times 17}$ are computed by conditioning on global posture embeddings:

$$\mathbf{A}_p = \text{softmax}(f_a(\mathbf{V}_{pg}) \times \mathbf{W} + \mathbf{b}), \quad (8)$$

where f_a is an FFN, and $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters.

Using the learned attention weights $\mathbf{A}_p$, the joint-wise distance features $\mathbf{V}_{pl,d}$ and orientation features $\mathbf{V}_{pl,a}$ are weighted accordingly and concatenated. The resulting representation is then projected through an FFN f_{pl} to obtain the local joint features $\mathbf{V}_{pl} \in \mathbb{R}^{N_p \times D_{pl}}$:

$$\mathbf{V}_{pl} = f_{pl}(\mathbf{V}_{pl,d} \cdot \mathbf{A}_p \oplus \mathbf{V}_{pl,a} \cdot \mathbf{A}_p). \quad (9)$$

(3) Feature fusion

Finally, we concatenate the coarse-grained global posture representation $\mathbf{V}_{pg}$ with the fine-grained local joint representation $\mathbf{V}_{pl}$ to form the final pose embedding $\mathbf{V}_{ps} \in \mathbb{R}^{N_p \times D_p}$, where $D_p = D_{pg} + D_{pl}$.

This fused representation preserves both holistic body configuration cues and discriminative joint-level geometry, enabling robust posture-aware interaction detection.

2. Enhanced Ablation Experiments on Posture-Related Components

To address the reviewer’s concern regarding insufficient ablation experiments on human body posture, we conducted comprehensive posture-related ablations at multiple levels.

First, we evaluated the contribution of the pose branch at the modality level by comparing different combinations of spatial, appearance, and pose cues (Table IV), which demonstrates that pose information provides consistent and complementary performance gains.

TABLE IV
THE ABLATION RESULTS (MAP $\times 100$) OF MODEL VARIANTS WITH DIFFERENT VISUAL CUES ON PHYSLAB AND HICO-DET TEST SETS.

#	Modality Fusion			PhysLab			HICO-DET		
	Spatial	Appearance	Pose	Full	Rare	Non-Rare	Full	Rare	Non-Rare
M1	✓			63.19	60.77	63.25	32.85	29.44	34.00
M2		✓		64.79	61.34	65.03	33.89	30.47	34.94
M3			✓	63.25	61.49	63.37	33.12	30.20	34.00
M4	✓	✓		65.27	62.33	65.48	34.03	30.59	35.09
M5	✓		✓	64.05	62.56	64.15	33.16	30.38	34.25
M6		✓	✓	65.72	63.90	65.85	34.30	30.98	35.29
M7	✓	✓	✓	66.34	64.22	66.49	34.58	31.25	35.69

Second, we performed component-level ablations within the pose branch by separately evaluating global posture encoding and fine-grained local joint geometry modeling (Table VI), verifying that fine-grained joint-level posture cues play a critical role in HOI recognition and are complementary to global posture features.

TABLE VI
THE ABLATION RESULTS (MAP $\times 100$) OF FEATURE COMPONENTS IN THE P-CR BRANCH ON PHYSLAB AND HICO-DET TEST SETS.

#	Features		PhysLab			HICO-DET		
	Global	Local	Full	Rare	Non-Rare	Full	Rare	Non-Rare
F1			65.27	62.33	65.48	34.29	30.59	35.39
F2	✓		65.80	63.62	65.95	34.42	30.43	35.61
F3		✓	66.07	63.73	66.23	34.47	30.81	35.62
F4	✓	✓	66.34	64.22	66.49	34.58	31.25	35.69

Third, we analyzed the effectiveness of the proposed Global Pose-Modulated Local Attention (GPLA) mechanism by comparing different attention configurations (Table V), confirming that pose-guided joint attention further enhances the utilization of fine-grained posture information.

TABLE V
THE ABLATION RESULTS (MAP $\times 100$) OF ATTENTION MECHANISMS ON PHYSLAB AND HICO-DET TEST SETS.

#	Attention		PhysLab			HICO-DET		
	A _a	A _p	Full	Rare	Non-Rare	Full	Rare	Non-Rare
H1			64.33	61.15	64.55	33.86	30.89	34.98
H2	✓		65.28	63.71	65.39	34.14	30.61	35.18
H3		✓	65.06	63.62	65.16	34.22	30.65	35.28
H4	✓	✓	66.34	64.22	66.49	34.58	31.25	35.69

Finally, we assessed the robustness of the pose branch by replacing the pose estimator with alternative models (e.g., OpenPose, AlphaPose, and HRNet) and different detector/backbone combinations (Table IX). The results indicate that the proposed posture modeling strategy is stable and generalizable across different pose estimation settings.

TABLE IX
COMPARISON OF DETECTION PERFORMANCE (MAP $\times 100$) WITH DIFFERENT CHOICES OF THE BACKBONE, OBJECT DETECTOR, AND POSE ESTIMATOR.

Fixed module	Compared module		PhysLab			HICO-DET		
			Full	Rare	Non-Rare	Full	Rare	Non-Rare
DETR+Openpose	+ResNet-50	46	66.34	64.22	66.49	34.58	31.25	35.69
	+ResNet-101	46	68.64	65.37	68.84	36.19	33.39	37.06
DETR+ResNet-50	+Openpose	33	66.34	64.22	66.49	34.58	31.25	35.69
	+AlphaPose	62	67.11	64.88	67.26	35.46	33.21	36.17
	+HRNet-W32	63	67.24	64.99	67.39	35.52	33.55	36.25
ResNet-50+Openpose	+DETR	30	66.34	64.22	66.49	34.58	31.25	35.69
	+D-DETR	64	67.70	64.79	67.90	35.29	31.63	36.44
	+H-DETR	48	68.21	65.34	68.41	37.32	32.14	38.85

Comment 2:

The CM-CQC module relies on the results of object detection and pose estimation in the early stage, but what is the impact of human pose estimation errors in situations such as occlusion and limb deformation on its subsequent interactive reasoning stage? Is there a relevant error correction mechanism to handle situations where detection uncertainty exists?

REPLY:

Thank you for this insightful comment. We agree that pose estimation errors caused by occlusion or severe limb deformation are a common and well-recognized challenge in two-stage HOI frameworks, and such errors may propagate to the subsequent interaction reasoning stage.

1. Impact of pose estimation errors and robustness.

- (1) In our framework, pose cues are not used as hard constraints but as soft contextual signals for interaction reasoning. The CM-CQC module jointly exploits appearance, spatial, and posture cues, where posture information further includes both global body configuration and local joint geometry, rather than relying on the precise location of individual keypoints. This multi-cue and global-to-local design provides a degree of robustness, allowing the model to rely on complementary visual evidence even when some joints are inaccurately estimated.
- (2) Moreover, we empirically evaluate the robustness of our framework by replacing upstream object detectors, backbones, and pose estimators with stronger alternatives (Table IX). The consistent performance gains indicate that our method is compatible with and can directly benefit from improvements in upstream detection and pose estimation.

TABLE IX
COMPARISON OF DETECTION PERFORMANCE (MAP $\times 100$) WITH DIFFERENT CHOICES OF THE BACKBONE, OBJECT DETECTOR, AND POSE ESTIMATOR.

Fixed module	Compared module	PhysLab			HICO-DET		
		Full	Rare	Non-Rare	Full	Rare	Non-Rare
DETR+Openpose	+ResNet-50	46	66.34	64.22	66.49	34.58	31.25
	+ResNet-101	46	68.64	65.37	68.84	36.19	33.39
DETR+ResNet-50	+Openpose	33	66.34	64.22	66.49	34.58	31.25
	+AlphaPose	62	67.11	64.88	67.26	35.46	33.21
	+HRNet-W32	63	67.24	64.99	67.39	35.52	33.55
ResNet-50+Openpose	+DETR	30	66.34	64.22	66.49	34.58	31.25
	+D-DETR	64	67.70	64.79	67.90	35.29	31.63
	+H-DETR	48	68.21	65.34	68.41	37.32	32.14

2. Error correction and scope of this work.

- (1) We agree with the reviewer that explicit mechanisms for handling detection uncertainty, such as end-to-end pose refinement or uncertainty-aware feedback loops, represent an important and meaningful research direction.
- (2) In the current work, we adopt a standard two-stage HOI framework, which is widely used in the literature. The advantages of this paradigm include stable and efficient training, clear task decoupling between perception and interaction reasoning, and strong modularity that allows plug-and-play integration of improved object detectors and pose estimators. However, a known limitation of such two-stage designs is that estimation errors from upstream modules (e.g., caused by occlusion or limb deformation) may propagate to downstream reasoning stages. This trade-off is inherent to the two-stage modeling paradigm and is shared by many existing HOI approaches.
- (3) We note that there exist related works specifically dedicated to modeling detection uncertainty or performing explicit error correction through end-to-end optimization or

uncertainty-aware mechanisms. While these methods are valuable and complementary, they focus on a different research setting and are not the core emphasis of our work, which aims at improving interaction reasoning given estimated visual cues.

- (4) We believe that uncertainty-aware reasoning and joint refinement constitute meaningful and promising research directions, and we plan to explore them in future work.

Comments 3:

The specific implementation process of the method proposed in the paper is not clear enough, and it is recommended to add pseudocode algorithms.

REPLY:

Thank you for the helpful suggestion. In the revised manuscript, we have added Algorithm 1 to explicitly describe the overall implementation pipeline of the proposed PM-EQC framework.

Below is our revised content [Algorithm 1 on page 3]:

reasoning. For completeness, Algorithm 1 provides a step-by-step description of the overall implementation pipeline.

Algorithm 1 The Proposed PM-EQC method

```

1: Input: image  $\mathbf{X}$ 
2: Output: HOI predictions  $\mathbf{R}$ 
3: // Stage A: Object detection branch (DETR-style)
4: Extract backbone features:  $\mathbf{V}_b \leftarrow \text{Backbone}(\mathbf{X})$ 
5: Compute positional embeddings  $\mathbf{P}$  according to Eqs. (1)–(3)
6: Encode image features:  $\mathbf{V}_e \leftarrow \text{Image Encoder}(\mathbf{V}_b, \mathbf{P})$ 
7: Random initialize instance queries  $\mathbf{Q}_{\text{ins}}$ 
8: Decode instance embeddings:  $\mathbf{V}_d \leftarrow \text{Instance Decoder}(\mathbf{V}_e, \mathbf{Q}_{\text{ins}})$ 
9: Predict classes and boxes of instances:  $\mathbf{C} \leftarrow f_c(\mathbf{V}_d)$ ,  $\mathbf{B} \leftarrow f_b(\mathbf{V}_d)$ 
10: Split detections into humans and objects:  $(\mathbf{C}_h, \mathbf{B}_h), (\mathbf{C}_o, \mathbf{B}_o) \leftarrow \text{Parse}(\mathbf{C}, \mathbf{B})$ 
11: // Stage B: Pose estimation branch
12: Estimate body keypoints and retain valid human poses:  $\mathbf{V}_p \leftarrow \text{Filter}(\text{Pose Estimator}(\mathbf{X}))$ 
13: // Stage C: Interaction Reasoning
14: Compose HOI queries from detected cues:  $\mathbf{Q}_{\text{hoi}} \leftarrow \text{CM-CQC}(\mathbf{V}_d, \mathbf{C}_h, \mathbf{B}_h, \mathbf{C}_o, \mathbf{B}_o, \mathbf{V}_p)$ 
15: Generate interaction embeddings  $\mathbf{V}_q$  with  $\mathbf{Q}_{\text{hoi}}$  and  $\mathbf{V}_b$  according to Eq. (4)
16: Predict interaction verb logits:  $\mathbf{C}_v \leftarrow f_v(\mathbf{V}_q)$ 
17: Form final HOI predictions:
18:  $\mathbf{R} \leftarrow \text{Assemble}(\mathbf{B}_h, \mathbf{B}_o, \mathbf{C}_o, \mathbf{C}_v)$ 
19: Return  $\mathbf{R}$ 

```

Comments 4:

The visualization analysis section of the paper on HOI mainly focuses on the performance of HOI in relatively simple scenarios, however, how does HOI perform in dense interaction and multi person interaction scenarios?

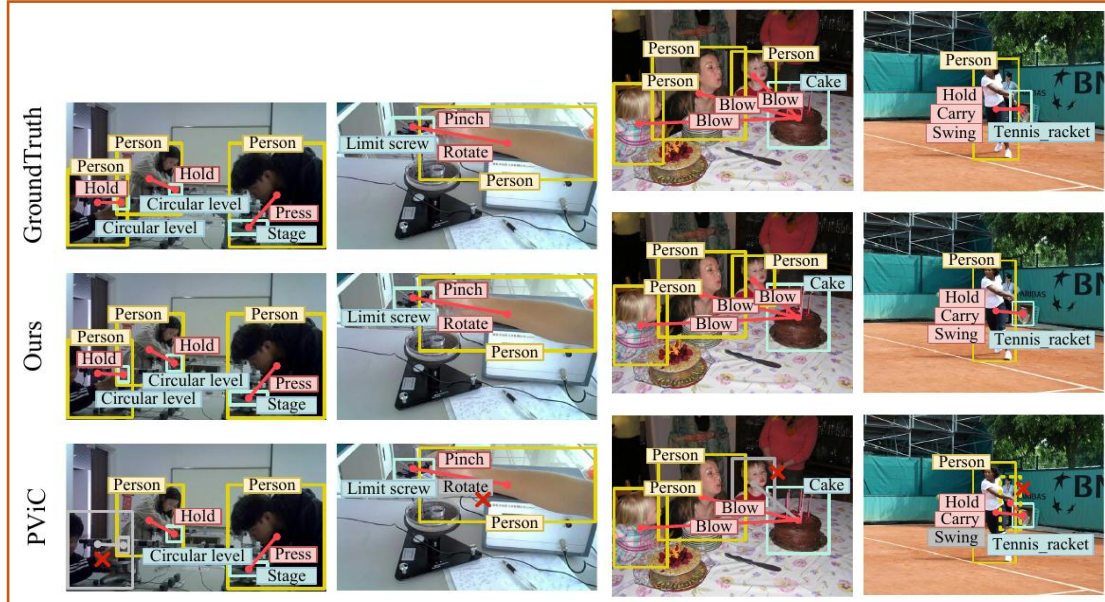
REPLY:

Thank you for this insightful comment. We agree that dense interaction and multi-person scenarios are important for evaluating HOI models.

We would like to clarify that, under the annotation protocol of HICO-DET and PhysLab,

although images may contain multiple humans and objects, the number of annotated HOIs per image is typically limited to one or two. As a result, extremely dense interaction graphs are rare in this benchmark.

Importantly, our existing visualizations already include representative multi-person and multi-interaction cases within this setting. As shown in figures below, these cases reflect the challenging interaction configurations available in the HICO-DET dataset.



Therefore, the presented examples already cover multi-person and multi-interaction scenarios as defined by the dataset annotations. We believe these visualizations are reasonably representative of dense and multi-person interaction cases within the scope of the benchmark, and help illustrate the behavior of our method under such settings.

Comments 5:

In addition, references 31 and 32 are duplicated, please correct them.

REPLY:

We thank the reviewer for the careful reading. The duplicated references (Refs. 31 and 32) have been identified and corrected in the revised manuscript.

Comments 6:

Suggest adding the relevant work, such as:

- [1] Locality-Aware Zero-Shot Human-Object Interaction Detection, in CVPR, 2025.
- [2] Object Centric Body Part Attention Network for Human-Object Interaction Detection, The 6th Chinese Conference on Pattern Recognition and Computer Vision, 2023. DOI: 10.1007/978-981-99-8555-5_30
- [3] ASCFormer: An Adaptive Structure-aware Cascaded Transformer for 3D Object Detection, IEEE Transactions on Circuits and Systems for Video Technology, 2025. DOI: 10.1109/TCSVT.2025.3612592

REPLY:

Thank you for the valuable suggestions. We appreciate the reviewer for pointing out these highly relevant works, which are closely related to our research topic.

Specifically, "Locality-Aware Zero-Shot Human-Object Interaction Detection" introduces a locality-aware modeling strategy to address zero-shot HOI detection by exploiting spatial priors and object-centric cues, demonstrating strong generalization to unseen interaction categories.

"Object-Centric Body Part Attention Network for Human-Object Interaction Detection" focuses on modeling fine-grained body-part relevance from an object-centric perspective, highlighting the importance of selective body-part attention for accurate HOI recognition.

In addition, "ASCFormer: An Adaptive Structure-aware Cascaded Transformer for 3D Object Detection" presents a structure-aware cascaded transformer architecture that effectively captures geometric and structural dependencies, which is conceptually related to our motivation of structured interaction reasoning.

We have carefully added and discussed these works in the revised Related Work section to better position our method with respect to recent advances. We believe that incorporating these references helps make the related work discussion more comprehensive and strengthens the contextual grounding of our contribution.

Below is our revised content:

Attention [31] over unary and pairwise markers [26], and pose-aware initialization [3], [21]. Although these designs

A key challenge in this paradigm lies in the fusion of heterogeneous features [25]. Existing works tackle this by exploring appearance and spatial cues [2], [26], [27], introducing

[21] Z. Liu and X. Zhang, "Object centric body part attention network for human-object interaction detection," in *Chinese Conference on Pattern Recognition and Computer Vision*, 2023, pp. 378–391.

[25] X. Zhang, X. Li, M. Zhou, M. Gan, and C. P. Chen, "ASCFormer: An adaptive structure-aware cascaded transformer for 3d object detection," *IEEE Transactions on Circuits and Systems for Video Technology*, 2025, Early Access.

[27] S. Kim, D. Jung, and M. Cho, "Locality-aware zero-shot human-object interaction detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 20 190–20 200.

Response to Reviewer 2

Comments:

All my concerns have been well addressed. I have no more comments.

REPLY:

We sincerely thank the reviewer for the positive assessment and careful evaluation of our manuscript. We are pleased that all concerns have been well addressed and appreciate the reviewer's time and effort.

Response to Reviewer 3

Comments 1:

Overall Assessment: The revised version of the paper demonstrates significant improvement in quality. The authors have comprehensively addressed the first-round review comments. The innovation (e.g., the hierarchical pose contextualization mechanism) and experimental completeness (covering the latest SOTA methods) meet the required standards. The paper structure

is clear, the code is publicly available, and the reproducibility is high. It is recommended to accept this paper.

REPLY:

We sincerely thank the reviewer for the positive evaluation and recommendation. We appreciate the recognition of the methodological contributions, experimental completeness, and reproducibility of our work.

Comments 2:

Summary of Resolved Issues : The authors have effectively resolved the initial concerns, including:

- ****Innovation****: Added comparisons with works from 2024-2025 (e.g., TED-Net, Zero-Shot HOI methods), highlighting the advantages of PM-EQC's visual modeling.
- ****Method Elaboration****: Clarified details of the CM-CQC module (e.g., the geometric mapping function G and BM threshold).
- ****Experiments****: Updated Tables I-III and Appendix A, enhancing timeliness and reproducibility.
- ****Writing Quality****: Corrected grammatical errors and unified the reference format.

REPLY:

We thank the reviewer for the positive summary and careful evaluation. We appreciate the recognition of the improvements in innovation, methodological clarity, experimental completeness, and writing quality, which have significantly strengthened the manuscript.

Comments 3:

New Existing Issues: Despite the substantial improvements in the revised version, the following new issues require attention:

1. ****Insufficient Explanation of Method Details****:

- The learning process of attention weights in the CM-CQC module (e.g., the optimization strategy for weight matrix W and bias b in Equation 8 of the GPLA mechanism) is not thoroughly explained, which may affect reproducibility. It is recommended to add descriptions of training dynamics in the methods section.
- For example, how are the parameters in Equation 8 adjusted via backpropagation? Additional context is needed.

REPLY:

Thank you for the reviewer's concern regarding the optimization of the parameters in the GPLA mechanism.

We would like to clarify that the projection parameters W and b are implemented as learnable parameters in PyTorch (i.e., `nn.Parameter`). Their functional role is equivalent to the weight and bias of a linear layer, and thus they are jointly optimized with the rest of the network via standard backpropagation.

Below is our revised content [\[the second to last paragraph of the first column on page 6\]](#):

where f_a is an FFN, and $W \in \mathbb{R}^{D_{ps} \times 17}$ and $b \in \mathbb{R}^{17}$ are learnable parameters, which are randomly initialized from a standard normal distribution and jointly optimized with the rest of the model via standard backpropagation.

2. ****Inadequate Depth in Experimental Analysis****:

- Failure case analysis for zero-shot settings (RF-UC/NF-UC) is missing; reliance on numerical results (e.g., Table III) may obscure model limitations. It is advised to include qualitative visualizations, such as confusion matrices or typical error examples, to evaluate generalization blind spots.

REPLY:

Thank you for the helpful suggestion.

Following this comment, we have added qualitative visualization results under the zero-shot setting in Fig. 8, together with a corresponding analysis of typical failure cases.

Below is our revised content [\[the second paragraph of Section IV.F on page 12 and Fig.8 on page 13\]](#):



- The computational cost analysis (Table X) lacks comparison with lightweight benchmarks (e.g., MobileNet-based methods), making it difficult to demonstrate efficiency advantages.

REPLY:

Thank you for your good suggestion regarding comparison with MobileNet-based methods.

We would like to note that, in the context of HICO-DET and related HOI benchmarks, there are currently very few public implementations or pretrained models built on lightweight backbones

such as MobileNet. This makes it difficult to compute and report model parameters and inference speed under a unified implementation, hardware, and evaluation protocol. In contrast, many recent competitive HOI methods are released with ResNet-based backbones and public implementations, which enables standardized and fair efficiency comparison. Moreover, among widely adopted high-performance HOI frameworks, ResNet already represents one of the more lightweight backbone configurations.

For practicality and fairness, our computational cost analysis therefore focuses on methods with publicly available implementations and ResNet-style backbones, which constitute the most commonly used and comparable settings in recent HOI literature (as also reflected in Tables I and II). While MobileNet is indeed more lightweight, the absence of open-source MobileNet-based HOI models makes direct empirical comparison difficult.

Method	Backbone	Method	Backbone
RCD ₃ [9]	ResNet-50	STIP [2]	ResNet-50
STIP [2]	ResNet-50	GEN-VLKT _s [41]	ResNet-50
GEN-VLKT _s [41]	ResNet-50	OCN [4]	ResNet-50
OCN [4]	ResNet-50	RLIP [24]	ResNet-50
RLIP [24]	ResNet-50	Wu et al. [6]	ResNet-50
Wu et al. [6]	ResNet-50	CDT [28]	ResNet-50
CDT [28]	ResNet-50	MUREN [19]	ResNet-50
MUREN [19]	ResNet-50	VT-HOI [42]	ResNet-50
VT-HOI [42]	ResNet-50	DiffHOI-S [14]	ResNet-50
DiffHOI-S [14]	ResNet-50	TED-Net [8]	ResNet-50
TED-Net [8]	ResNet-50	PDN-M [43]	ResNet-50
PDN-M [43]	ResNet-50	SOV-STG-S [7]	ResNet-50
SOV-STG-S [7]	ResNet-50	KI2HOI [10]	ResNet-50
KI2HOI [10]	ResNet-50	PM-EQC-S (Ours)	ResNet-50
PM-EQC-S (Ours)	ResNet-50	OCN [4]	ResNet-101
OCN [4]	ResNet-101	UPT [26]	ResNet-101
UPT [26]	ResNet-101	GEN-VLKT _l [41]	ResNet-101
GEN-VLKT _l [41]	ResNet-101	CQL+GEN-VLKT _l [44]	ResNet-101
CQL+GEN-VLKT _l [44]	ResNet-101	OpenCat [45]	ResNet-101
OpenCat [45]	ResNet-101	PDN-L [43]	ResNet-101
PDN-L [43]	ResNet-101	SOV-STG-L [7]	ResNet-101
SOV-STG-L [7]	ResNet-101	PM-EQC-M (Ours)	ResNet-101
PM-EQC-M (Ours)	ResNet-101	OCN [4]	ResNet-101
OCN [4]	ResNet-101	UPT [26]	ResNet-101
UPT [26]	ResNet-101	GEN-VLKT _l [41]	ResNet-101
GEN-VLKT _l [41]	ResNet-101	SOV-STG-L [7]	ResNet-101
SOV-STG-L [7]	ResNet-101	PM-EQC-M (Ours)	ResNet-101
PM-EQC-M (Ours)	ResNet-101	ERNet [22]	ENetV2-XL
ERNet [22]	ENetV2-XL	PViC [5]	Swin-L
PViC [5]	Swin-L	FGAHOI [15]	Swin-L
FGAHOI [15]	Swin-L	VC-HOI [11]	ViT-B/16
VC-HOI [11]	ViT-B/16	SOV-STG-Swin-L [7]	Swin-L
SOV-STG-Swin-L [7]	Swin-L	TKCE [12]	ResNet-50+ViT-L
TKCE [12]	ResNet-50+ViT-L	PM-EQC-L (Ours)	Swin-L
PM-EQC-L (Ours)	Swin-L		

Nevertheless, following the reviewer’s suggestion, we conducted an additional lightweight reference experiment by replacing the DETR detector in UPT with SSDLite-MobileNetV3-L. We selected UPT for this study because most existing one-stage HOI methods tightly couple interaction reasoning with intermediate DETR features, making detector replacement impractical. In contrast, the two-stage design of UPT decouples object detection from interaction inference, which allows for a more controlled detector substitution.

As shown in the Table X, the MobileNet-based variant significantly reduces inference time (from 98.46 ms to 64.37 ms), but also leads to a substantial drop in HOI detection performance (from 65.28 mAP to 57.01 mAP).

TABLE X
COMPARISON OF COMPUTATIONAL COST ON PE-HOI WITH NVIDIA
Geforce RTX 3090.

Type	Method	Detector	mAP $\uparrow$	Params (A/T) $\downarrow$	Time $\downarrow$
One-stage	QPIC [1]	DETR+ResNet-50	54.07	41.68/41.46M	39.13ms
	OCN [4]	DETR+ResNet-50	52.19	43.77/43.55M	47.42ms
	CDN [17]	DETR+ResNet-50	59.44	41.67/41.45M	39.95ms
	MUREN [19]	DETR+ResNet-50	63.27	75.19/74.95M	99.85ms
Two-stage	STIP [2]	DETR+ResNet-50	62.62	54.71/13.19M	71.38ms
	UPT [26]	DETR+ResNet-101	65.28	73.70/31.31M	98.46ms
	UPT [26]	SSDLite+MobileNetV3-L	57.01	35.75/31.31M	64.37ms
	PM-EQC-S (Ours)	DETR+ResNet-50	66.34	96.46/30.70M	97.84ms

However, for HOI detection, accuracy remains the primary evaluation criterion in existing benchmarks and most application scenarios, as interaction reasoning critically relies on high-quality human-object detections. Under this setting, the performance degradation introduced by the lightweight detector is often more detrimental than the corresponding speed gain. Moreover, the MobileNet-based UPT variant is inferior to the one-stage method CDN in both inference speed and detection accuracy. In comparison, although our method incurs a modest increase in inference time, it achieves a more favorable overall trade-off in detection accuracy and training complexity (i.e., the number of trainable parameters).

Since this MobileNet-based variant is not an official HOI baseline and involves a manual modification of the detector architecture, we report it here solely as a reference rather than including it as a primary comparison in the manuscript. We believe that this supplementary result helps clarify the efficiency–accuracy trade-off and further contextualize the advantages of our method, without compromising the fairness or consistency of the main experimental comparisons.

3. **Minor Writing Issues**:

- Some paragraphs are redundant (e.g., Section II on related work) and could be further condensed to improve readability.

REPLY:

Thank you for the helpful comment. Following this suggestion, we have carefully revised Section II (Related Work) by removing redundant descriptions, condensing overlapping discussions, and improving the overall organization to enhance readability, while preserving the key technical context and representative references.

Below is our revised content [Section II on pages 2-3]:

A. One-stage Methods

One-stage methods have emerged as the mainstream paradigm in HOI detection owing to their end-to-end design and high inference efficiency. Early works typically relied on Transformer-based architectures to jointly localize Human-Object (HO) pairs and predict interactions, implicitly modeling pairwise relations through attention mechanisms [1], [7], [14]–[18]. However, most of these methods employ a shared predictor for both instance localization and interaction recognition, thereby limiting the decoupling of instance-level reasoning from relation-level modeling.

To alleviate this limitation, recent studies introduce architectural refinements such as decoupled decoder branches [19]–[21], multi-scale feature fusion [15], and deformable attention mechanisms [22]. Some works further incorporate external semantic signals, including pseudo-labels and caption-derived priors, to guide query learning and reduce ambiguity [14], [23], [24]. While effective, these text-driven designs often increase annotation costs and weaken visual grounding, potentially limiting scalability in practical scenarios.

B. Two-stage Methods

In contrast, two-stage approaches adopt a decoupled pipeline, where humans and objects are first detected by a pretrained detector, followed by interaction inference based on fixed instance regions [2], [3], [6], [7]. This formulation narrows the search space for interaction reasoning and often leads to improved accuracy and reduced training complexity.

A key challenge in this paradigm lies in the fusion of heterogeneous features [25]. Existing works tackle this by exploring appearance and spatial cues [2], [26], [27], introducing pose-aware attention [3], and utilizing fine-grained body-part representations [6]. Some approaches further introduce semantic embeddings of objects and verbs to enrich category-level representations [4], [7]. Despite these advances, how to effectively represent and integrate diverse cues for fine-grained interaction understanding remains an open problem.

C. Query Learning

Queries play a central role in Transformer-based HOI detectors by serving as interaction carriers in the decoder [3], [19], [22], [26]. One-stage approaches typically initialize queries randomly [15], [19]. While computationally simple, such initialization lacks informative priors, resulting in limited expressiveness. To mitigate this issue, prior works have explored several directions, including decoder-level dependency modeling [22], [28], text-based supervision for query enhancement [14], [24], [28], and representation enrichment via distillation or deeper architectures [23], [29], [30].

In contrast, two-stage methods generally initialize queries from object detector outputs [2], [3], [5], [26], thereby narrowing the search space and providing stronger priors for interaction reasoning. Representative strategies include spatially structured queries for HO pairs [2], DETR-style Self-Attention [31] over unary and pairwise markers [26], and pose-aware initialization [3], [21]. Although these designs improve query initialization, challenges remain in capturing fine-grained interaction cues and integrating heterogeneous context, highlighting the need for more expressive query construction mechanisms.

- There are slight inconsistencies in reference formatting (e.g., inconsistent abbreviation of titles), requiring final proofreading.

REPLY:

Thank you for pointing this out. We have carefully proofread the manuscript and corrected inconsistencies in reference formatting, including title abbreviations, to ensure consistency with the journal's style guidelines.

For example, we corrected the 'Qpic' to 'QPIC' to align with the official title of the paper.

Concluding Remarks

We sincerely thank the **Editor-in-Chief, Associate Editor, Senior Area Editor, and all Reviewers** for your valuable time and insightful comments.

We have carefully addressed every suggestion and revised the manuscript to improve its technical clarity, presentation quality, and overall completeness. Your feedback has been highly constructive and has greatly strengthened the work.

We appreciate your thoughtful evaluation and hope that the revised version meets your expectations.